

## Introduction

we will study about factorization which has an important role in mathematics. It helps us to reduce the complicated expression into simple expressions.

### 4.1 Factorization

Let  $p(x)$ ,  $q(x)$  and  $r(x)$  are three polynomials such that,  $p(x) \times q(x) = r(x)$ . Here, the resulting polynomial  $r(x)$  is the product of  $p(x)$  and  $q(x)$ , and the polynomials  $p(x)$  and  $q(x)$  are called the **factors** of  $r(x)$ .

There are some examples of factors of the polynomials are given below.

- (i)  $6x^2y^3 = (2 \times 3)(x \times x)(y \times y \times y)$
- (ii)  $ax + aby + abcz = a(x + by + bcz)$
- (iii)  $5x + 15xy = 5x(1 + 3y)$
- (iv)  $x - y = (\sqrt{x})^2 - (\sqrt{y})^2 = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})$

#### 4.1.1 Recall Factorization of Expressions of the Following Types

- (i)  $ka + kb + kc$  (Common factors in all the terms)
- (ii)  $ac + ad + bc + bd$  (Common factors in pairs of terms)
- (iii)  $a^2 \pm 2ab + b^2$  (Perfect squares)
- (iv)  $a^2 - b^2$  (Difference of two squares)
- (v)  $a^2 \pm 2ab + b^2 - c^2$
- (vi)  $(\sqrt{a})^2 - (\sqrt{b})^2$

#### (i) Factors of the type: $ka + kb + kc$

Let us see the following examples

**Example 01** Factorize:  $10a + 15b - 20c$

**Solution:**  $10a + 15b - 20c$   
 $= 5(2a + 3b - 4c)$  (5 as a common from the expression)

**Example 02** Find the factors of  $\frac{4}{9}x - \frac{8}{12}x - \frac{16}{15}xy$

**Solution:**  $\frac{4}{9}x - \frac{8}{12}x - \frac{16}{15}xy$   
 $= \frac{4}{3 \cdot 3}x - \frac{4 \cdot 2}{3 \cdot 4}x - \frac{4 \cdot 4}{3 \cdot 5}xy$   
 $= \frac{4}{3} \left( \frac{1}{3}x - \frac{2}{4}x - \frac{4}{5}xy \right)$  (Taking  $\frac{4}{3}$  as a common from the expression)

**(ii) Factors of the type:  $ac + ad + bc + bd$** 

See the following examples

**Example 01** Find the factors of  $3a - ac - 3c + c^2$ 

$$\begin{aligned}
 \text{Solution: } & 3a - ac - 3c + c^2 \\
 &= a(3 - c) - c(3 - c) \\
 &= (3 - c)(a - c)
 \end{aligned}$$

**Example 02**  $9y^2z + 3xyz - 9xy^2 - 3x^2y$ 

$$\begin{aligned}
 \text{Solution: } & 9y^2z + 3xyz - 9xy^2 - 3x^2y \\
 &= 3yz(3y + x) - 3xy(3y + x) \\
 &= (3y + x)(3yz - 3xy) \\
 &= (3y + x) \times 3y(z - x) \\
 &= 3y(x + 3y)(z - x) \text{ are the required factors.}
 \end{aligned}$$

**(iii) Factors of the type:  $a^2 \pm 2ab + b^2$** 

As we know that,

$$\begin{aligned}
 a^2 + 2ab + b^2 &= (a)^2 + 2(a)(b) + (b)^2 = (a + b)^2 \\
 \text{and } a^2 - 2ab + b^2 &= (a)^2 - 2(a)(b) + (b)^2 = (a - b)^2
 \end{aligned}$$

Let us see the following examples

**Example 01** Factorize  $16a^2 + 40ab + 25b^2$ 

$$\begin{aligned}
 \text{Solution: } & 16a^2 + 40ab + 25b^2 \\
 &= (4a)^2 + 2(4a)(5b) + (5b)^2 \\
 &= (4a + 5b)^2 \qquad [\because a^2 + 2ab + b^2 = (a + b)^2]
 \end{aligned}$$

**Example 02** Factorize  $4p^2 - 28pq + 49q^2$ 

$$\begin{aligned}
 \text{Solution: } & 4p^2 - 28pq + 49q^2 \\
 &= (2p)^2 - 2(2p)(7q) + (7q)^2 \\
 &= (2p - 7q)^2 \qquad [\because a^2 - 2ab + b^2 = (a - b)^2]
 \end{aligned}$$

**(iv) Factors of the type:  $a^2 - b^2$** 

Let us see following examples

**Example 01** Factorize  $4x^2 - 1$ 

$$\begin{aligned}
 \text{Solution: } & 4x^2 - 1 \\
 &= (2x)^2 - (1)^2 \\
 &= (2x - 1)(2x + 1) \qquad [\because a^2 - b^2 = (a - b)(a + b)]
 \end{aligned}$$



**Example 02** Factorize  $96y^2 - 6z^2$ 

**Solution:**

$$\begin{aligned}
 & 96y^2 - 6z^2 \\
 &= 6(16y^2 - z^2) \\
 &= 6[(4y)^2 - z^2] \\
 &= 6(4y - z)(4y + z) \qquad [\because a^2 - b^2 = (a - b)(a + b)]
 \end{aligned}$$

**Example 03** Factorize  $9r^4 - (6s - t^2)^2$ 

**Solution:**

$$\begin{aligned}
 & 9r^4 - (6s - t^2)^2 \\
 &= (3r^2)^2 - (6s - t^2)^2 \\
 &= [3r^2 - (6s - t^2)][3r^2 + (6s - t^2)], \qquad [\because a^2 - b^2 = (a - b)(a + b)] \\
 &= (3r^2 - 6s + t^2)(3r^2 + 6s - t^2)
 \end{aligned}$$

**(v) Factors of the type:  $(a^2 \pm 2ab + b^2) - c^2$** 

Let us see following examples.

**Example 01** Factorize  $x^2 + 4xy^2 + 4y^4 - 4z^2$ 

**Solution:**

$$\begin{aligned}
 & x^2 + 4xy^2 + 4y^4 - 4z^2 \\
 &= \{(x)^2 + 2(x)(2y^2) + (2y^2)^2\} - (2z)^2 \\
 &= (x + 2y^2)^2 - (2z)^2 \\
 &= \{(x + 2y^2) + 2z\}\{(x + 2y^2) - 2z\} \qquad [\because a^2 - b^2 = (a - b)(a + b)] \\
 &= (x + 2y^2 + 2z)(x + 2y^2 - 2z)
 \end{aligned}$$

**Example 02**  $9p^2 - 6pq + q^2 - 9r^2$ 

**Solution:**

$$\begin{aligned}
 & 9p^2 - 6pq + q^2 - 9r^2 \\
 &= (3p)^2 - 2(3p)(q) + q^2 - (3r)^2 \\
 &= (3p - q)^2 - (3r)^2 \\
 &= (3p - q + 3r)(3p - q - 3r) \qquad [\because a^2 - b^2 = (a - b)(a + b)]
 \end{aligned}$$

**(vi) Factors of the type:  $(\sqrt{a})^2 - (\sqrt{b})^2$** **Example 01** Factorize  $(\sqrt{xy})^2 - (\sqrt{z})^2$ 

**Solution:**

$$\begin{aligned}
 & (\sqrt{xy})^2 - (\sqrt{z})^2 \\
 &= (\sqrt{xy} - \sqrt{z})(\sqrt{xy} + \sqrt{z}) \qquad [\because a^2 - b^2 = (a - b)(a + b)]
 \end{aligned}$$

## Exercise 4.1

## 1. Factorize the following:

(i)  $4x + 16y + 24z$

(ii)  $x^2 + 3x^2y + 4x^2y^2z$

(iii)  $3pqr + 6pqt + 3pqs$

(iv)  $9qr(s^2 + t^2) + 18q^2r^2(s^2 + t^2)$

(v)  $\frac{z^2x}{16} - \frac{x^2z^2}{8} + \frac{x^2z^3}{12}$

(vi)  $a(x-y) - a^2b(x-y) + a^2b^2(x-y)$

## 2. Factorize the following:

(i)  $7x + xz + 7z + z^2$

(ii)  $9a^2b + 18ab^2 - 6ac - 12bc$

(iii)  $6t - 12p + 4tq - 8pq$

(iv)  $r^2 + 9rs - 7rs - 63s^2$

(v)  $\frac{y^2}{4} - \frac{y^2z}{4} - \frac{z^2t}{9} + \frac{z^3t}{9}$

(vi)  $\frac{10xy}{11} + \frac{5xz}{11} - \frac{14y^2}{11} - \frac{7yz}{11}$

## 3. Factorize:

(i)  $4a^2 + 12ab + 9b^2$

(ii)  $36x^4 + 12x^2 + 1$

(iii)  $x^2 + 1 + \frac{1}{4x^2}$

(iv)  $81y^2 + 144yz + 64z^2$

(v)  $625 + 50a^2b + a^4b^2$

(vi)  $a^2 + 0.4a + 0.04$

## 4. Factorize:

(i)  $b^4 - 4b^2c^2 + 4c^4$

(ii)  $\frac{9}{4}x^4 - 2 + \frac{4}{9x^4}$

(iii)  $2a^3b^3 - 16a^2b^4 + 32ab^5$

(iv)  $9(p+q)^2 - 6(p+q)r^2 + r^4$

(v)  $x^2y^2 - 0.1xy + 0.0025$

(vi)  $(a-b)^2 - 18(a-b) + 81$

## 5. Factorize:

(i)  $4a^2 - 9b^2$

(ii)  $16x^2 - 25y^2$

(iii)  $100x^2z^2 - y^4$

(iv)  $\frac{1}{100}x^4 - 100y^4$

(v)  $\frac{64}{81}f^2 - \frac{81}{64}g^4$

(vi)  $\frac{x^4}{121} - 121y^2$

## 6. Factorize:

(i)  $(2x+z)^2 - (2x-z)^2$

(ii)  $(4a-9b)^2 - (2a+5b)^2$

(iii)  $169x^4 - (3t+4)^2$

(iv)  $(9x^2 - 4y^2)^2 - (4x^2 - y^2)^2$

(v)  $\left(a^2 + 2 + \frac{1}{a^2}\right) - \left(b^2 - 2 + \frac{1}{b^2}\right)$

(vi)  $9x^2 + \frac{1}{9x^2} - 4y^2 - \frac{1}{4y^2} + 4$

## 7. Factorize:

(i)  $(x^2 + 2xy + y^2) - 9z^4$

(ii)  $(4a^2 + 8ab^2 + 4b^4) - 9c^2$





(iii)  $16d^4 - (c^4 - 2c^2d + d^2)$

(v)  $x^2 - y^2 - 4x - 2y + 3$

(iv)  $4(x^2 + 2xy^2 + y^4) - 9y^6$

(vi)  $4x^2 - y^2 - 2y - 1$

8. **Factorize:**

(i)  $(\sqrt{ab})^2 - (\sqrt{c})^2$

(ii)  $(\sqrt{4x})^2 - (\sqrt{9y})^2$

(iii)  $(\sqrt{yz})^2 - \left(\frac{1}{\sqrt{yz}}\right)^2$

(iv)  $xzt - \frac{1}{t}$

**4.1.2 Factorize the expression of following types.**Type I:  $a^4 + a^2b^2 + b^4$  or  $a^4 + 4b^4$ 

This type includes those algebraic expressions which are neither perfect squares nor in the form of the difference of two squares. Factorization of this type is explained in the following examples.

**Example 01** Factorize:  $a^4 + a^2b^2 + b^4$ .

**Solution:**

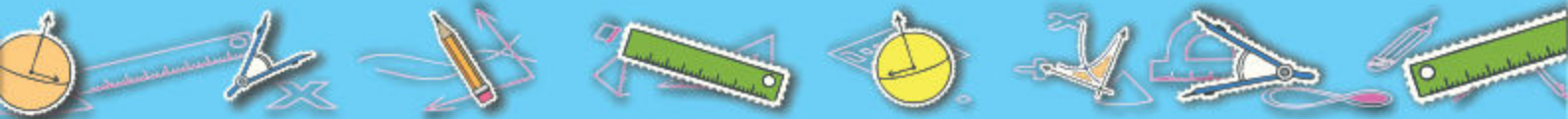
$$\begin{aligned}
 & a^4 + a^2b^2 + b^4 \\
 &= (a^4 + b^4) + a^2b^2 && \text{(Rearrange the terms)} \\
 &= (a^4 + 2a^2b^2 + b^4) - 2a^2b^2 + a^2b^2 && [\text{by adding and subtracting } 2a^2b^2] \\
 &= \{(a^2)^2 + 2(a^2)(b^2) + (b^2)^2\} - a^2b^2 \\
 &= (a^2 + b^2)^2 - (ab)^2 \\
 &= \{(a^2 + b^2) - ab\}\{(a^2 + b^2) + ab\} && [\because a^2 - b^2 = (a - b)(a + b)] \\
 &= (a^2 - ab + b^2)(a^2 + ab + b^2)
 \end{aligned}$$

**Example 02** Factorize:  $a^4 + 4b^4$ 

**Solution:**

$$\begin{aligned}
 & a^4 + 4b^4 \\
 &= (a^2)^2 + (2b^2)^2 \\
 &= (a^2)^2 + (2b^2)^2 + 2(a^2)(2b^2) - 2(a^2)(2b^2) \\
 &= \{(a^2)^2 + 2(a^2)(2b^2) + (2b^2)^2\} - 4a^2b^2 \\
 &= (a^2 + 2b^2)^2 - (2ab)^2 \\
 &= \{(a^2 + 2b^2) - 2ab\}\{(a^2 + 2b^2) + 2ab\} && [\because a^2 - b^2 = (a - b)(a + b)] \\
 &= (a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2)
 \end{aligned}$$

[For completing the square adding and subtracting  $2(a^2)(2b^2)$ ]



**Example 03** Factorize:  $x^8 + x^4 + 1$ 

**Solution:**

$$\begin{aligned}
 & x^8 + x^4 + 1 \\
 &= (x^8 + 1) + x^4 \\
 &= \{(x^4)^2 + (1)^2 + 2(x^4)(1)\} - 2(x^4)(1) + x^4 \\
 &= (x^4 + 1)^2 - x^4 \\
 &= (x^4 + 1)^2 - (x^2)^2 \\
 &= \{(x^4 + 1) - x^2\}\{(x^4 + 1) + x^2\} \quad [\because a^2 - b^2 = (a - b)(a + b)] \\
 &= \{(x^4 + x^2 + 1)\}(x^4 - x^2 + 1) \\
 &= \{(x^2 + 1)^2 - 2x^2 + x^2\}(x^4 - x^2 + 1) \\
 &= \{(x^2 + 1)^2 - x^2\}(x^4 - x^2 + 1) \\
 &= (x^2 + x + 1)(x^2 - x + 1)(x^4 - x^2 + 1)
 \end{aligned}$$

[For completing the square adding and subtracting  $2(x^4)(1)$ ]

**Type II:**  $x^2 + px + q$ 

This type of expression can be factorized by breaking the middle term process.

**Example 01** Factorize  $y^2 + 7y + 12$ 

**Solution:**

$$\begin{aligned}
 & y^2 + 7y + 12 \\
 &= y^2 + 3y + 4y + 12 \\
 &= y(y + 3) + 4(y + 3) \\
 &= (y + 3)(y + 4)
 \end{aligned}$$

**Example 02** Factorize  $x^2 + 13xy - 30y^2$ 

**Solution:**

$$\begin{aligned}
 & x^2 + 13xy - 30y^2 \\
 &= x^2 + 15xy - 2xy - 30y^2 \\
 &= x(x + 15) - 2y(x + 15) \\
 &= (x + 15)(x - 2y)
 \end{aligned}$$

**Type III:**  $ax^2 + bx + c, a \neq 0$ .

To factorize the expression  $ax^2 + bx + c, a \neq 0$ , the following steps are needed:

- Find the product  $ac$ , where  $a$  is coefficient of  $x^2$  and  $c$  is constant.
- Find two numbers  $x_1$  and  $x_2$  such that  $x_1 + x_2 = b$  and  $x_1 x_2 = ac$ .

To explain this method the following examples are helpful.



**Example 01** Factorize  $10x^2 - 19xy + 6y^2$ 

**Solution:**

$$\begin{aligned} & 10x^2 - 19xy + 6y^2 \\ &= 10x^2 - 15xy - 4xy + 6y^2 \\ &= 5x(2x - 3y) - 2y(2x - 3y) \\ &= (2x - 3y)(5x - 2y) \end{aligned}$$

**Example 02** Factorize  $4x^2 + 12x + 5$ 

**Solution:**

$$\begin{aligned} & 4x^2 + 12x + 5 \\ &= 4x^2 + 10x + 2x + 5 \\ &= 2x(2x + 5) + 1(2x + 5) \\ &= (2x + 5)(2x + 1) \end{aligned}$$

**Example 03** Factorize  $3x^2 + 4x - 4$ 

**Solution:**

$$\begin{aligned} & 3x^2 + 4x - 4 \\ &= 3x^2 + 6x - 2x - 4 \\ &= 3x(x + 2) - 2(x + 2) \\ &= (x + 2)(3x - 2) \end{aligned}$$

**Example 04** Factorize  $6x^2 - x - 7$ 

**Solution:**

$$\begin{aligned} & 6x^2 - x - 7 \\ &= 6x^2 - 7x + 6x - 7 \\ &= x(6x - 7) + 1(6x - 7) \\ &= (6x - 7)(x + 1) \end{aligned}$$

### Exercise 4.2

1. Factorize the following:

(i)  $a^4 + a^2x^2 + x^4$

(iii)  $a^8 + a^4x^4 + x^8$

(ii)  $b^4 + b^2 + 1$

(iv)  $z^8 + z^4 + 1$

2. Factorize:

(i)  $x^4 + 4y^4$

(iii)  $4t^4 + 625$

(ii)  $36x^4z^4 + 9y^4$

(iv)  $4t^4 + 1$

3. Resolve into factors:

(i)  $x^2 + 3x - 10$

(iii)  $y^2 + 7y - 98$

(ii)  $a^2b^2 - 3ab - 10$

(iv)  $x^2y^2z^2 + 2xyz - 24$

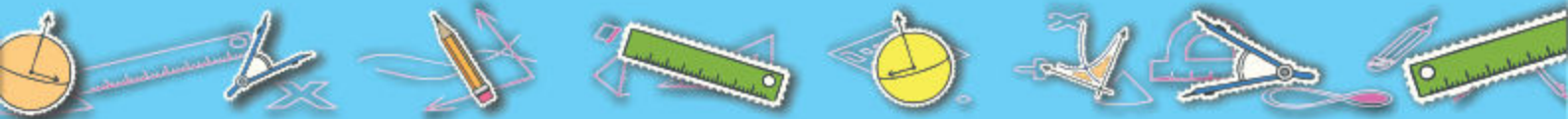
4. Resolve into factors:

(i)  $9y^2 + 21yz - 8z^2$

(iii)  $4x^2 + 12x + 5$

(ii)  $42x^2 - 8x - 2$

(iv)  $3x^2 - 38xy - 13y^2$



**Type IV:**  $(ax^2 + bx + c)(ax^2 + bx + d) + k$ 

$$(x+a)(x+b)(x+c)(x+d) + k$$

$$(x+a)(x+b)(x+c)(x+d) + kx^2$$

We shall explain the procedure of factorizing of these types expressions with the help of following examples.

**Example 01** Factorize:  $(x^2+5x+4)(x^2+5x+6) - 120$ **Solution:**  $(x^2+5x+4)(x^2+5x+6) - 120$ Let  $x^2+5x = t$ , then we have

$$\begin{aligned}
 & (t+4)(t+6) - 120 \\
 &= t^2 + 10t + 24 - 120 \\
 &= t^2 + 10t - 96 \\
 &= t^2 - 6t + 16t - 96 && \text{(by factorizing)} \\
 &= t(t-6) + 16(t-6) \\
 &= (t-6)(t+16) \\
 &= (x^2+5x-6)(x^2+5x+16) && [\because t=x^2+5x] \\
 &= (x^2-x+6x-6)(x^2+5x+16) \\
 &= [x(x-1)+6(x-1)](x^2+5x+16) \\
 &= (x-1)(x+6)(x^2+5x+16)
 \end{aligned}$$

**Example 02** Factorize:  $(x+1)(x+2)(x+3)(x+4) - 15$ **Solution:**  $(x+1)(x+2)(x+3)(x+4) - 15$ 

$$\text{Here } 1+4 = 2+3 = 5$$

$$\begin{aligned}
 & (x+1)(x+4)(x+2)(x+3) - 15 \quad \text{by arranging the factors} \\
 &= (x+1)(x+4)(x+2)(x+3) - 15
 \end{aligned}$$





$$\begin{aligned}
 &= (x^2+5x+4)(x^2+5x+6) - 15 \\
 &= (t+4)(t+6) - 15 \quad \text{where } t = x^2+5x \\
 &= t^2+10t+24-15 \\
 &= t^2+10t+9 \\
 &= (t+1)(t+9) \\
 &= (x^2+5x+1)(x^2+5x+9) \quad \because t = x^2+5x
 \end{aligned}$$

**Example 03** Factorize:  $(x+2)(x-2)(x-3)(x+3)+(-2x^2)$

**Solution:**

$$\begin{aligned}
 &(x+2)(x-2)(x-3)(x+3)+(-2x^2) \\
 &= (x+2)(x-2)(x-3)(x+3)+(-2x^2) \\
 &= (x^2-2^2)(x^2-3^2)-2x^2 \quad [\because (a+b)(a-b)=a^2-b^2] \\
 &= (x^2-4)(x^2-9)-2x^2 \\
 &= x^4-9x^2-4x^2+36-2x^2 \\
 &= x^4-15x^2+36 \\
 &= x^4-3x^2-12x^2+36 \\
 &= x^2(x^2-3)-12(x^2-3) \\
 &= (x^2-3)(x^2-12) \\
 &= [(x)^2-(\sqrt{3})^2][(x)^2-(2\sqrt{3})^2] \\
 &= (x-\sqrt{3})(x+\sqrt{3})(x-2\sqrt{3})(x+2\sqrt{3})
 \end{aligned}$$

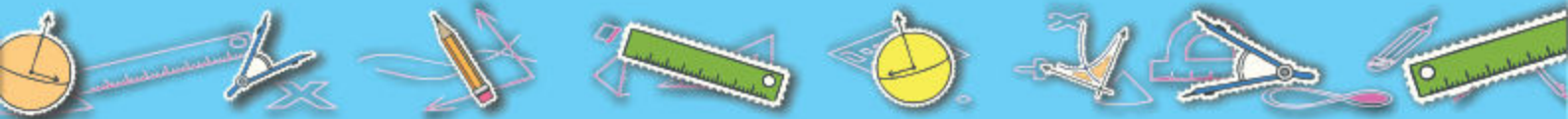
### Exercise 4.3

1. Factorize the following:

- |                                 |                                 |
|---------------------------------|---------------------------------|
| (i) $(x^2-4x-5)(x^2-4x-12)-144$ | (ii) $(x^2+5x+6)(x^2+5x+4)-3$   |
| (iii) $(x^2-2x+3)(x^2-2x+4)-42$ | (iv) $(x^2-8x+4)(x^2-8x-4)+15$  |
| (v) $(x^2+9x-1)(x^2+9x+5)-7$    | (vi) $(x^2-5x+4)(x^2-5x+6)-120$ |

2. Factorize:

- |                                 |                                 |
|---------------------------------|---------------------------------|
| (i) $(x+1)(x+2)(x+3)(x+4)-48$   | (ii) $(x+2)(x+3)(x+4)(x+5)-24$  |
| (iii) $(x-1)(x-2)(x-3)(x-4)-99$ | (iv) $(x-3)(x-5)(x-7)(x-9)+15$  |
| (v) $(x-1)(x-2)(x-3)(x-4)-224$  | (vi) $(x-2)(x-3)(x-4)(x-5)-255$ |



## 3. Factorize

(i)  $(x-2)(x-3)(x+2)(x+3)-2x^2$

(ii)  $(x-1)(x+1)(x+3)(x-3)-3x^2-23$

(iii)  $(x-1)(x+1)(x-3)(x+3)+4x^2$

(iv)  $(x-2)(x+2)(x-4)(x+4)-14x^2$

(v)  $(x+5)(x+2)(x-5)(x-2)+4x^2$

(vi)  $(x^2-x-12)(x^2-x-12)-x^2$

**Type V:**  $a^3+3a^2b+3ab^2+b^3$  and  $a^3-3a^2b+3ab^2-b^3$ 

As we know that

$$a^3+3a^2b+3ab^2+b^3=(a+b)^3$$

and

$$a^3-3a^2b+3ab^2-b^3=(a-b)^3.$$

The following examples will help us to understand the factorization of the types mentioned above.

**Example 01** Factorize (i)  $8x^3+12x^2y+6xy^2+y^3$  (ii)  $64x^3-12x^2+\frac{3x}{4}-\frac{1}{64}$ 

**Solution (i):**  $8x^3+12x^2y+6xy^2+y^3$   
 $= (2x)^3+3(2x)^2(y)+3(2x)(y)^2+(y)^3$  [ $\because a^3+3a^2b+3ab^2+b^3=(a+b)^3$ ]  
 $= (2x+y)^3$

**Solution (ii):**  $64x^3-12x^2+\frac{3x}{4}-\frac{1}{64}$   
 $= (4x)^3-3(4x)^2\left(\frac{1}{4}\right)+3(4x)\left(\frac{1}{4}\right)^2-\left(\frac{1}{4}\right)^3$  [ $\because a^3-3a^2b+3ab^2-b^3=(a-b)^3$ ]  
 $= \left(4x-\frac{1}{4}\right)^3$



## Exercise 4.4

## 1. Factorize the following:

(i)  $b^3 + 3b^2c + 3bc^2 + c^3$

(ii)  $8x^3 + 12x^2y + 6xy^2 + y^3$

(iii)  $64x^3 + 12x^2 + \frac{3x}{4} + \frac{1}{64}$

(iv)  $8x^3 + 36x^2 + 54x + 27$

(v)  $\frac{1}{27} + \frac{1}{3}y^2 + y^4 + y^6$

(vi)  $\frac{8}{27}x^3 + 2x^2y + \frac{9}{2}xy + \frac{27}{8}y^3$

(vii)  $\frac{64}{27} + \frac{16}{3}x + 4x^2 + x^3$

(viii)  $\frac{z^3}{8} + \frac{z^2y}{4} + \frac{zy^2}{6} + \frac{y^3}{27}$

## 2. Factorize:

(i)  $d^3 - 6d^2c + 12dc^2 - 8c^3$

(ii)  $x^6 + \frac{16}{3}x^2 - 4x^4 - \frac{64}{27}$

(iii)  $\frac{x^3}{125} - \frac{3}{25}x^2y + \frac{3}{5}xy^2 - y^3$

(iv)  $125z^3 - 75z^2y^2 + 15zy^4 - y^6$

(v)  $\frac{z^3}{27} - 2z^2y + 36zy^2 - 216y^3$

(vi)  $\frac{b^6}{27} - \frac{b^4c^2}{6} + \frac{b^2c^4}{4} - \frac{c^6}{8}$

(vii)  $216 + \frac{9z^2}{2} - 54z - \frac{z^3}{8}$

(viii)  $\frac{8}{27}x^3 - 2x^2y + \frac{9}{2}xy^2 - \frac{27}{8}y^3$

Type VI:  $a^3 \pm b^3$ 

As we know that,

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2),$$

$$\text{and } a^3 - b^3 = (a-b)(a^2 + ab + b^2).$$

The following example will help us to understand the factorization of above mentioned type.

**Example 01** Factorize  $8x^3 + 27$ **Solution:**

$$8x^3 + 27$$

$$= (2x)^3 + (3)^3$$

$$= (2x+3)[(2x)^2 - (2x)(3) + (3)^2] \quad [\because a^3 + b^3 = (a+b)(a^2 - ab + b^2)]$$

$$= (2x+3)(4x^2 - 6x + 9)$$

$$\text{Therefore } 8x^3 + 27 = (2x+3)(4x^2 - 6x + 9)$$

**Example 02** factors of  $108x^4 - 256xz^3$ .

**Solution:**

$$\begin{aligned}
 &108x^4 - 256xz^3 \\
 &= 4x(27x^3 - 64z^3) \\
 &= 4x[(3x)^3 - (4z)^3] \quad [\because a^3 - b^3 = (a-b)(a^2 + ab + b^2)] \\
 &= 4x(3x - 4z)[(3x)^2 + (3x)(4z) + (4z)^2] \\
 &= 4x(3x - 4z)(9x^2 + 12xz + 16z^2)
 \end{aligned}$$

Therefore:  $108x^3 - 256xz^3 = 4x(3x - 4z)(9x^2 + 12xz + 16z^2)$ .

**Note:** For complete factorization of the expression of type  $x^n - y^n$ , where  $n = 6k, \forall k \in \mathbb{N}$ ; apply difference of two squares formula first, then apply difference of two cubes formula.

**Exercise 4.5****1. Factorize the following:**

|                      |                                          |                         |                                         |
|----------------------|------------------------------------------|-------------------------|-----------------------------------------|
| (i) $x^3 + 8y^3$     | (ii) $a^{11} + a^2b^9$                   | (iii) $a^6 + 1$         | (iv) $a^3b^3 + 512$                     |
| (v) $a^3b^3 + 27b^6$ | (vi) $\frac{x^3}{125} + \frac{125}{x^3}$ | (vii) $x^9 + x^3y^6z^9$ | (viii) $\frac{x^6}{27} + \frac{8}{x^3}$ |

**2. Factorize**

|                                          |                        |                                   |                               |
|------------------------------------------|------------------------|-----------------------------------|-------------------------------|
| (i) $x^3 - 8y^3$                         | (ii) $x^9 - 8y^9$      | (iii) $1000 - \frac{x^3y^3}{125}$ | (iv) $a^6 - b^6$              |
| (v) $\frac{x^6}{64} - \frac{64}{x^{12}}$ | (vi) $x^{12} - y^{12}$ | (vii) $\frac{27}{x^3} - 8y^6$     | (viii) $8x^6 - \frac{1}{729}$ |

**4.2 Remainder and Factor Theorems**

Remainder and factor theorems are usually used to find the factors of the polynomial expressions of third and higher degrees of the polynomials.

**4.2.1 State and prove Remainder Theorem and Explain through examples****Statement:**

When a polynomial  $p(x)$  of degree  $n \geq 1$  is divided by a linear polynomial  $(x-a)$ , then remainder  $R$  is obtained by putting  $x=a$  i.e.  $R = p(a)$ . We can write  $p(x)$  as  $p(x) = q(x)(x-a) + R$ , (which is called division algorithm) where  $R$  is a constant (remainder), and the degree of  $q(x)$  is less than the degree of  $p(x)$  by 1.



**Proof:**

By division algorithm

$$p(x) = q(x)(x-a) + R \quad \dots\dots\dots (i)$$

As (i) is an identity, so it is true for all values of  $x$ . $\therefore$  Putting  $x = a$  in (i), we get

$$p(a) = q(a) \times (a-a) + R,$$

$$p(a) = q(a) \times 0 + R$$

 $\Rightarrow p(a) = R = \text{remainder. Hence Proved.}$ **4.2.2 Find remainder (without dividing) when a polynomial is divided by a linear polynomial.**

The following examples help us to use of remainder theorem.

**Example 01** Find the remainder when  $x^2 - 3x + 4$  is divided by  $x - 2$ **Solution:** Let  $p(x) = x^2 - 3x + 4$ Here  $a = 2$  by remainder theorem

$$p(2) = (2)^2 - 3(2) + 4$$

$$= 4 - 6 + 4 = 8 - 6 = 2$$

$$p(2) = R = 2$$

Thus, the remainder is 2.

**Example 02** Find the value of  $k$ , if the polynomial  $x^3 + kx^2 + 3x - 4$  leaves a remainder  $-2$  when divided by  $x + 2$ .**Solution:**Here  $p(x) = x^3 + kx^2 + 3x - 4$ 

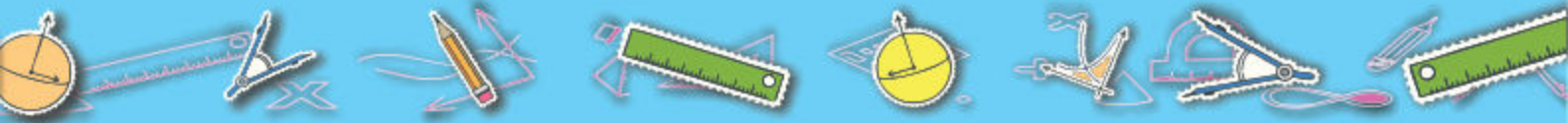
$$\therefore p(-2) = (-2)^3 + k(-2)^2 + 3(-2) - 4$$

$$\Rightarrow -2 = 4k - 18 \quad [\because \text{Remainder} = -2 = p(-2)]$$

$$\Rightarrow 4k = -2 + 18$$

$$\Rightarrow 4k = 16$$

$$\Rightarrow k = 4,$$

Thus, the value of  $k$  is 4.**4.2.3 Zero of a polynomial.**Let  $p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$  be a polynomial with real coefficient.By putting  $x = a$  in the polynomial  $p(x)$ , if the value of  $p(x)$  becomes zero i.e.  $p(a) = 0$ . Then ' $a$ ' is called the zero of polynomial  $p(x)$ .**Example 01** If  $p(x) = x + 7$  then  $-7$  is a zero of polynomial as  $p(-7) = -7 + 7 = 0$ .



**4.2.4 State and prove factor theorem.****Statement:**

The linear polynomial  $x-a$  is a factor of the polynomial  $p(x)$  iff,  $p(a) = 0$ .

**Proof:**

Let  $q(x)$  be the quotient and  $R$  be the remainder when a polynomial  $p(x)$  is divided by  $x-a$

Then by division algorithm, we have

$$p(x) = (x-a)q(x) + R \quad \dots\dots\dots (i)$$

By the Remainder theorem,

$R = p(a)$ , using in (i), we get

Thus, 
$$p(x) = q(x)(x-a) + p(a)$$

It is given that  $p(a)=0$ , then  $p(x)=q(x)(x-a)$ .

Note that  $p(x)$  is expressed as a product of  $q(x)$  and  $(x-a)$ .

Thus,  $(x-a)$  is a factor of the polynomial  $p(x)$ . Hence proved.

The following examples will help us to use of factor theorem.

**Example 01** Determine whether  $x+2$  is a factor of  $x^3 + \frac{9}{2}x^2 + 3x - 4$  or not.

**Solution:** Let  $p(x) = x^3 + \frac{9x^2}{2} + 3x - 4$

putting  $x = -2$ , we get

$$\begin{aligned} \therefore R = p(-2) &= (-2)^3 + \frac{9}{2}(-2)^2 + 3(-2) - 4 \\ &= -8 + 18 - 6 - 4 \\ &= -18 + 18 = 0 \Rightarrow R = 0, \end{aligned}$$

Remainder = 0,

$\therefore x+2$  is factor of  $p(x)$

**Example 02** Determine whether  $x+3$  is a factor of Thus,  $x^3 - x^2 - 8x + 12$

**Solution:**

$$\text{Let } p(x) = x^3 - x^2 - 8x + 12$$

$\therefore$  putting  $x = -3$ , we get

$$\begin{aligned} R = p(-3) &= (-3)^3 - (-3)^2 - 8(-3) + 12 \\ &= -27 - 9 + 24 + 12 \end{aligned}$$

$$R = p(-3) = -36 + 36 = 0$$

As remainder is found to be zero

$\therefore x+3$  is factor of  $x^3 - x^2 - 8x + 12$

## Exercise 4.6

- Find the remainder by using the remainder theorem when
  - $x^3 - 6x^2 + 11x - 8$  is divided by  $(x-1)$
  - $x^3 + 6x^2 + 11x + 8$  is divided by  $(x+1)$
  - $x^3 - x^2 + 14$  is divided by  $(x-2)$
  - $x^3 - 3x^2 + 4x - 14$  is divided by  $(x+2)$
  - $(2y-1)^3 + 6(3+4y) - 9$  is divided by  $(2y+1)$
  - $4y^3 - 4y + 3$  is divided by  $(2y-1)$
  - $(2y+1)^3 - 6(3-4y) - 10$  is divided by  $(2y-1)$
  - $x^4 + x^2y^2 + y^4$  is divided by  $(x-y)$ .
- Find the value of  $m$ , if  $p(y) = my^3 + 4y^2 + 3y - 4$  and  $q(y) = y^3 - 4y + m$  leaves the same remainder when divided by  $(y-3)$ .
- If the polynomial  $4x^3 - 7x^2 + 6x - 3k$  is exactly divisible by  $(x+2)$ , find the value of  $k$ .
- Find the value of  $r$ , if  $(y+2)$  is a factor of the polynomial  $3y^2 - 4ry - 4r^2$ .

## 4.3 Synthetic Division

Synthetic division is a method to divide a polynomial by a linear polynomial.

## 4.3.1 Describe the method of synthetic division.

The method of synthetic division is described with the help of following example.

**Example 1.** Divide the polynomial  $p(x) = x^3 - 3x^2 + 5x + 7$  by  $(x-1)$  using synthetic division

**Solution:** Here  $x-1=0 \Rightarrow x=1$ , (1 is a multiplier).

Write down the coefficients of the given polynomial.

Thus,

|   |   |    |    |    |         |
|---|---|----|----|----|---------|
| 1 | 1 | -3 | 5  | 7  | (Row 1) |
|   |   | 1  | -2 | 3  | (Row 2) |
|   | 1 | -2 | 3  | 10 | (Row 3) |
|   |   |    |    | R  |         |



**Description**

- Step I.** In Row 1, write the coefficients and constant term of the given polynomial  $p(x)$  in the descending order. If any term is missing in  $p(x)$  then insert zero for that term.
- Step II.** Write the first coefficient in Row 3, below its position in Row 1.
- Step III.** Write the product of 1 (the multiplier) and the coefficient (1) in the Row 3 beneath the 2<sup>nd</sup> coefficient in Row 2, and add, putting the sum below them in the Row 3 and so on.

Thus,  $q(x)=x^2-2x+3$  and  $p(1) = R = 10$ .

**Note:** Degree of  $q(x)=[\text{Degree of } p(x)]-1=3-1=2$   
and the last element of the row 3, is the remainder.

**4.3.2 Use of Synthetic division to:**

- Find quotient and remainder when a given polynomial is divided by a linear polynomial.
- Find the value(s) of unknown(s) if zeros of the polynomial are given.
- Find the value(s) of unknown(s) if the factors of the polynomial are given.

**Example 01** Find quotient and remainder when,  
 $p(x) = x^4 - 12x^3 + 50x^2 - 84x + 49$  is divided by a linear polynomial  $x-5$

**Solution:** Given that  $p(x)=x^4-12x^3+50x^2-84x+49$ ,  
and linear polynomial  $x-a = x-5$ , i.e.,  $a=5$  is the multiplier = 5  
 $p(5)=R=?$  and  $q(x)=?$

To find quotient and the remainder, we will use synthetic division method as under.

|   |   |     |     |     |       |         |
|---|---|-----|-----|-----|-------|---------|
| 5 | 1 | -12 | 50  | -84 | 49    | (Row 1) |
|   |   | 5   | -35 | 75  | -45   | (Row 2) |
|   | 1 | -7  | 15  | -9  | 4 = R | (Row 3) |

Thus,  $q(x)=x^3-7x^2+15x-9$  and  $R=4$  are the required quotient and remainder respectively.

**Note:**  $R \neq 0$ , therefore  $(x-5)$  is not the factor of given polynomial  $p(x)$ .

**Example 02** For what value of  $m$ , 1 is a zero of the polynomial

$$p(x) = x^3 - mx^2 + x - 1$$

**Solution:** Given that,

$$p(x) = x^3 - mx^2 + x - 1$$

Here, the multiplier  $a = 1$ .

By synthetic division method we have,

|   |   |       |       |           |         |
|---|---|-------|-------|-----------|---------|
| 1 | 1 | $-m$  | 1     | $-1$      | (Row 1) |
|   |   | 1     | $1-m$ | $2-m$     | (Row 2) |
|   | 1 | $1-m$ | $2-m$ | $1-m = R$ | (Row 3) |

Since 1 is the zero of  $p(x)$ , so  $R = 0$

$$\Rightarrow 1 - m = 0$$

$$\Rightarrow m = 1$$

Thus, for 1 as a zero of the polynomial  $p(x)$   $m$  must be equal to 1.

### Exercise 4.7

1. By using synthetic division method to divide the following polynomials and also find their quotient and remainder.

(i)  $p(x) = x^3 - x^2 + x - 1$  by  $x - 1$

(ii)  $p(x) = x^3 - x^2 - x - 1$  by  $x + 1$

(iii)  $p(x) = x^3 - 6x^2 + 11x - 6$  by  $x + 2$

(iv)  $p(x) = x^3 + 6x^2 - 11x - 6$  by  $x - 2$

(v)  $p(x) = x^4 - x^3 + x^2 - x - 1$  by  $x + 2$

(vi)  $p(x) = x^4 + x^3 - x^2 + x - 1$  by  $x - 1$

(vii)  $p(x) = x^5 + x^3 - 2x^2 - 3$  by  $x + 3$

(viii)  $p(x) = x^5 - x^4 + x^3 - 3x^2 + 6x - 6$  by  $x - 3$

(ix)  $p(x) = 2x^4 - 2x^3 + 100x^2 - 168x + 95$  by  $x - 2$

(x)  $p(x) = 6x^4 - 72x^3 + 300x^2 - 564x + 270$  by  $x - 5$

2. For what value of  $k$ ,  $-2$  is zero of the polynomial  $p(x) = x^3 + x^2 - 14x - k$

3. For what value of  $m$ ,  $(x - 2)$  be factor of  $x^3 + mx^2 - 7x - 10$

4. For what value of  $m$ ,  $(x + 2)$  is factor of  $4x^3 - 7x^2 + 6x - 3m$

5. For what value of  $m$ ,  $-1$  is a zero of the polynomial,  $p(x) = 2x^3 - 4mx^2 + x - 1$



## 4.4 Factorization of Cubic Polynomial

We have already studied the method of solving linear and quadratic polynomials. Now we will find the factors of cubic polynomials using factor theorem.

### 4.4.1 Use factor theorem to factorize a cubic polynomial

To factorize a cubic polynomial by factor theorem, it is necessary that one of the factor or more of the zeros of the polynomial is (are) known.

Let us see the following examples:

**Example 01** Factorize  $x^3 - 6x^2 + 11x - 6$

**Solution:** Let  $p(x) = x^3 - 6x^2 + 11x - 6$   
The factors of 6 are  $\pm 1, \pm 2, \pm 3, \pm 6$

$$p(1) = (1)^3 - 6(1)^2 + 11(1) - 6$$

$$p(1) = 1 - 6 + 11 - 6$$

$$p(1) = 0$$

Hence,  $x-1$  is a factor of  $p(x)$

By synthetic division

|   |   |    |    |    |         |
|---|---|----|----|----|---------|
| 1 | 1 | -6 | 11 | -6 | (Row 1) |
|   |   | 1  | -5 | 6  | (Row 2) |
|   | 1 | -5 | 6  | 0  | (Row 3) |

By division algorithm,  $p(x) = (x-a)q(x) + R$

$$\begin{aligned} \therefore p(x) &= (x-1)(x^2 - 5x + 6) + 0 \\ &= (x-1)\{x^2 - 2x - 3x + 6\} \\ &= (x-1)\{x(x-2) - 3(x-2)\} \\ p(x) &= (x-1)(x-2)(x-3) \end{aligned}$$

**Example 02** Factorize  $x^3 - 4x^2 + x + 6$

**Solution:** The factors of 6 are  $\pm 1, \pm 2, \pm 3, \pm 6$

If  $x-1$  is a factor of  $p(x)$

Then,

$$\begin{aligned} p(1) &= (1)^3 - 4(1)^2 + 1 + 6 \\ &= 1 - 4 + 1 + 6 \\ &= 4 \neq 0 \end{aligned}$$

Hence,  $x-1$  is not factor of  $p(x)$ .

If  $x+1$  is a factor of  $p(x)$

Then,

$$\begin{aligned} p(-1) &= (-1)^3 - 4(-1)^2 + 1(-1) + 6 \\ &= -1 - 4 - 1 + 6 \\ &= 0 \end{aligned}$$

Hence,  $x+1$  is factor of  $p(x)$ .

By synthetic division

|      |   |      |   |      |         |
|------|---|------|---|------|---------|
| $-1$ | 1 | $-4$ | 1 | 6    | (Row 1) |
|      |   | $-1$ | 5 | $-6$ | (Row 2) |
|      | 1 | $-5$ | 6 | 0    | (Row 3) |

By division algorithm,  $p(x) = (x-a)q(x) + R$

$$\begin{aligned} \therefore p(x) &= (x+1)(x^2 - 5x + 6) + 0 \\ &= (x+1)\{x^2 - 2x - 3x + 6\} \\ &= (x+1)\{x(x-2) - 3(x-2)\} \\ p(x) &= (x+1)(x-2)(x-3) \end{aligned}$$

### Exercise 4.8

**Factorize by using factor theorem**

- |                            |                            |                             |
|----------------------------|----------------------------|-----------------------------|
| 1. $x^3 - x^2 + x - 1$     | 2. $x^3 + x^2 - x - 1$     | 3. $x^3 - 6x^2 + 11x - 6$   |
| 4. $x^3 + 5x^2 - 4x - 20$  | 5. $x^3 - 2x^2 + 9x - 18$  | 6. $6x^3 + 7x^2 - x - 2$    |
| 7. $x^3 + 8x^2 + 19x + 12$ | 8. $2x^3 + 9x^2 + 10x + 3$ | 9. $x^3 + 12x^2 + 44x + 48$ |

### Review Exercise 4

#### 1. True and false questions

Read the following sentences carefully and encircle 'T' in case of true and 'F' in case of false statement.

- |                                       |     |
|---------------------------------------|-----|
| (i) $x^2 + x - 6 = (x+3)(x-2)$        | T/F |
| (ii) $a^3 + 27 = (a+3)(a^2 - 3a + 9)$ | T/F |
| (iii) $b^3 - 8 = (b-2)(b^2 + 2b + 4)$ | T/F |



(iv)  $a^4 - b^4 = (a - b)(a + b)(a + b)^2$  T/F

(v)  $a^6 + b^6 = (a^3 + b^3)(a^3 - b^3)$  T/F

(vi)  $a^5 + b^5 = (a + b)^5$  T/F

**Complete the following sentences**

(i)  $16x^2 - y^4 = (4x - y^2)$  \_\_\_\_\_

(ii)  $x^3 - 64y^3 = (x - 4y)$  \_\_\_\_\_

(iii)  $x^2 + 5x + 6 = (x + 2)$  \_\_\_\_\_

(iv)  $x^2 + y^2 = (x - y)^2$  \_\_\_\_\_

(v)  $a^3 + 27b^3 = (a + 3b)$  \_\_\_\_\_

**Tick (✓) the correct answers**(i) Factors of  $a^2 + 2a - 24$  are:

(a)  $a + 4, a - 6$

(b)  $a - 4, a + 6$

(c)  $a + 3, a - 8$

(d)  $a + 8, a - 3$

(ii) Factors of  $a^2 + 2ab + b^2 - c^2$  is:

(a)  $(a - b + c)(x - b - c)$

(b)  $(a + b + c)(a - b - c)$

(c)  $(a + b + c)(a + b - c)$

(d)  $(a + b + c)(a - b - c)$

(iii) Factors of  $x^3 + y^3$  is:

(a)  $(x - y)(x^2 + xy + y^2)$

(b)  $(x + y)(x^2 + xy + y^2)$

(c)  $(x + y)(x^2 + xy - y^2)$

(d)  $(x + y)(x^2 - xy + y^2)$

(iv) Factors of  $y^3 - 27z^3$  are:

(a)  $y - 3z, y^2 + 3yz + 9z^2$

(b)  $y - 3z, y + 3z + 9z^2$

(c)  $y - 3z, y^2 - 3yz + 9z^2$

(d)  $y + 3z, y^2 - 3yz + 9z^2$

(v) In simplified form  $\frac{1}{x + y} + \frac{y}{x^2 - y^2} =$ 

(a)  $\frac{y + 1}{x^2 - y^2}$

(b)  $\frac{x}{x^2 - y^2}$

(c)  $\frac{y}{x^2 - y^2}$

(d)  $\frac{y - 1}{x^2 - y^2}$

(vi) Find  $m$ , so that  $x^2 + 4x + m$  is a complete square

(a) 16

(b) -16

(c) 4

(d) -4



## Summary

- Factorization is the process in which we express the given polynomial (expression) as a product of two or more expressions.
- We learn and resolve into factors of the following types of formulas:

$$\begin{aligned} \text{(i)} \quad & ka + kb + kc = k(a + b + c) \\ \text{(ii)} \quad & \underline{ac + ad} + \underline{bc + bd} = a(c + d) + b(c + d) = (a + b)(c + d) \\ \text{(iii)} \quad & a^2 + 2ab + b^2 = (a)^2 + 2(a)(b) + (b)^2 = (a + b)^2 \\ \text{(iv)} \quad & a^2 - 2ab + b^2 = (a)^2 - 2(a)(b) + (b)^2 = (a - b)^2 \\ \text{(v)} \quad & a^2 - b^2 = (a - b)(a + b) \end{aligned}$$

- Remainder and factor theorems are two important theorems; these are used to find the factors of such type of polynomials which cannot be solved by the given formulas:

$$\begin{aligned} \text{(i)} \quad & a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3 & \text{(ii)} \quad & a^3 - 3a^2b + 3ab^2 - b^3 = (a - b)^3 \\ \text{(iii)} \quad & a^3 + b^3 = (a + b)(a^2 - ab + b^2) & \text{(iv)} \quad & a^3 - b^3 = (a - b)(a^2 + ab + b^2). \end{aligned}$$

- Zeros of the Polynomial**

If specific number  $x=a$  is substituted for the variable ' $x$ ' in a polynomial  $p(x)$  such that, the value of  $p(a)$  is zero, then ' $a$ ' is called a zero of the Polynomial  $p(x)$ .

- Factor theorem** can also be stated as:

The linear polynomial  $(x - a)$  is a factor of the polynomial  $p(x)$  iff,  $p(a) = 0$

- Description of synthetic division method**

**Step I.** Write in Row 1, the coefficients of  $p(x)$  in the descending powers of  $x$ . If any term is missing in  $p(x)$ , then insert zero for that term.

**Step II.** Write the first coefficient in Row below its position in Row 1.

- Step III.** Write the product of 2 multiplier and this coefficient in the Row 2 beneath the 2<sup>nd</sup> coefficient in Row 1, and added, putting the sum below them in the Row 3 and so on.



Unit

5

• Weightage = 5%

# ALGEBRAIC MANIPULATION

## Student Learning Outcomes (SLOs)

After completing this unit, students will be able to:

- ◆ Find Highest Common Factor (H.C.F) and Least Common Multiple (L.C.M) of algebraic expressions by factorization method.
- ◆ Know the relationship between H.C.F and L.C.M
- ◆ Use division method to determine highest common factor and least common multiple.
- ◆ Solve real life problems related to HCF and LCM
- ◆ Use highest common factor and least common multiple to reduce fractional expressions involving addition, subtraction, multiplication and division.
- ◆ Find square root of an algebraic expression by factorization and division methods.
- ◆ Solve real life problems related to HCF and LCM.